Project Name	Randomized trial to study the effect of rifaximin on gut microbiome diversity post allogeneic stem cell transplant in acute leukemia.		**CTRI/2023/07/054803** **Principal Investigator: Dr Anant Gokarn** **(Contact: anantgokarn@gmail.com)** **Primary Sponsor: Lady Tata Memorial Trust**
Description	Nanopore 16S sequencing of human stool sample collected at baseline time point		
Sequencing Type	Full-length V1-V9 bacterial 16S gene amplicon		
Nucleic Acid Purification	**DNA Extraction Kit**	Promega Fecal Microbial Kit & Promega Maxwell RSC instrument (Promega Corporation, WI, USA)	
	Cell Lysis Mechanism	Chemical, Enzymatic, Mechanical bead-beating	Lysis buffer & Proteinase K (Promega), Zirconium beads 1.0-1.2 mm (Neuation), bead-beating at 3000 RPM for 1 min, cooling for 1 min for 3 cycles
	DNA Purification Mechanism	Robotic Magnetic bead-based (Promega)	
Library Prep	**Kit/Protocol Name**	SQK-16S114.24 (Oxford Nanopore Tech, UK)	Rapid sequencing DNA - 16S Barcoding Kit 24 V14 MinION protocol
	PCR Master Mix	LongAmp HotStart Taq 2X (NEB)	
	Other/Notes	If not amplified with LongAmp, repeat PCR with repliQa HiFi ToughMix master mix (QuantaBio)	
Sequencing	**Technology**	Nanopore	Dorado Basecalling
	Device	MinION Mk1B	
	Flow Cell Type & Chemistry	MinION FLO-MIN114, R10.4.1	**Raw output format:** POD5 **Base called data format:** FASTQ
Bioinformatics	**Bioinformatic Pipeline**	Emu KD Curry et al, *Nature Methods* 2022. https://github.com/arpit20328/Nextflow-Pipeline-for-16S-Nanopore-Data-Analysis	Min abundance: 0.0001 Alignment Algorithm: Minimap2 (with Default parameters) Reference Database used: RDB + NCBI (built by EMU Pipeline)

	Min Q score	6	High Accuracy Base calling with Reads having Barcodes at Both Sides (Perfect Amplicons) The Nanopore Dorado model used for base calling: dna_r10.4.1_e8.2_400bps_hac@v5.0.0
	Quality Score Filtering Tool Used	NanoFilt	
Control Experiment Benchmark	**Zymobiomics Community Standard II (Zymo Research)**	We got 5 Species out of 8 Mentioned in the theoretical abundance matrix. A detailed Control Report is mentioned in the SOP for this study. Alpha Diversity: 5 Inverse Simpson's Index (ISI) 1.0825 Shannon Diversity Index (H): 0.208278	**Benchmark Parameters Used:** 1. **The quality score of 6 with High Accuracy Basecalling** 2. **ISI Error Rate of** -0.5303 3. **H Error Rate of** +1.36% **Theoretical ISI : 1.088** **Theoretical H: 0.205**

Table C: Summary of the wet lab procedure.